

SHAW CAMPBELL

Website | campbellshaw8@gmail.com | [LinkedIn](#) | [GitHub](#) | Cell: 0797336179

Graduate software developer with a BSc Eng (Hons) in Electrical & Computer Engineering from UCT.
Experienced software developer with machine learning research experience and a good academic track record.

WORK EXPERIENCE



Graduate Software Developer — PSG Financial Services

2026 – Present

Building and maintaining production web applications, working across a .NET / Angular stack. Write end-to-end tests using Playwright in C#.

Skills: [C#](#) · [.NET](#) · [ASP.NET](#) · [Angular \(Typescript, CSS\)](#) · [Agile](#) · [Web Development](#) · [MS SQL Server](#)



Software Engineering Vacation Work — Peralex Electronics

2025 · 5 weeks

Designed and built a Python desktop application for visualising and navigating openEMS electromagnetic simulation results across multiple antenna types. Participated in the full Agile workflow — sprint planning, stand-ups, and peer code reviews.

Skills: [Python](#) · [OOP](#) · [Desktop App Development](#) · [Agile](#) · [Electromagnetics](#)

Reference: [Skippy Burger](#), skippy@peralex.com, [reference letter](#)



Software Development Intern — Electrum Payments

2024 · 4 weeks

Delivered a real feature on a live company codebase, embedded within the engineering team across stand-ups, planning sessions, and code reviews. Received a full-time offer at the conclusion of the internship.

Java · [Spring Boot](#) · [OOP](#) · [REST APIs](#) · [Agile](#)

Reference: [Skippy Burger](#), [Johan Du Toit](#), [reference letter](#)



Signals and Systems II Tutor (EEE3092F)

Jan – Jul 2025 · 4 months

Signals and Systems I Tutor (EEE2047F)

Aug – Nov 2025 · 4 months

Control Systems Laboratory Assistant

2023 & 2024 · 6 weeks per year

Computer Science Tutor — Python (CSC1015F)

2023 · 6 months

Computer Science Tutor — Java (CSC1016S)

2023 · 6 months

SELECTED PROJECTS

Deep Learning Deformable Registration of Ultrasound to MRI for Prostate Imaging

[Thesis](#) · [2025](#)

[GitHub](#)

Final-year honours thesis investigating weakly supervised deep learning for deformable registration of transrectal ultrasound (TRUS) and MRI in prostate imaging.

[Python](#) · [PyTorch](#) · [VoxelMorph](#) · [ANTs](#) · [Jupyter Notebooks](#) · [Medical Image Analysis](#)

Micro-Mouse — Sensing & Power Subsystems

[Embedded](#) · [2024](#)

[GitHub](#)

Designed and implemented the infrared sensor subsystem and power subsystem for an autonomous maze-solving micro-mouse, interfacing with an existing motherboard and dual-motor drive.

[C](#) · [STM32L476](#) · [KiCad](#) · [Embedded Systems](#) · [PCB Design](#) · [Electronics](#)

TaskFlow — CPU-Inspired Task Scheduler

[C++](#) · [2025](#)

[GitHub](#)

A back-end C++ task scheduler that adapts classical CPU scheduling algorithms for human productivity planning.

[C++](#) · [OOP](#) · [Design Patterns](#) · [Scheduling Algorithms](#) · [CMake](#)

Self Studying Stochastic Calculus

[Stochastic Calculus](#) · [2026](#)

Out of love for the rigorous and abstract theory, I have been self studying real analysis and stochastic calculus using textbooks from UCT's masters of financial engineering programme. Progress, handwritten exercise solutions, and a dated study log are available on my [website](#).

EDUCATION



BSc Eng (Hons) — Electrical & Computer Engineering

University of Cape Town · 2021 – 2025



Honours Graduate

GPA = 71% · Thesis in medical ML



Dean's Merit List x2

2022 & 2023



PSG Full Merit Bursary

2024 & 2025



UCT Plus Gold Award

60 hrs community service · 2023

EXTRA-CURRICULARS

• Student Residence Mentor

Jan 2022 – Nov 2023 & Jan 2024 – Nov 2024

Supported first-year students through the academic and social transition into university life across two separate years.

• IT & Maintenance Representative — Leo Marquard Hall House Committee

Jan 2023 – Nov 2023

Built and maintained a website for the campus residence. Provided IT support across the residence community.

TECH I HAVE WORKED WITH QUITE A LOT

[Python](#) · [Java](#) · [C++](#) · [C](#) · [C#](#) · [Linux](#) · [.NET](#) · [ASP.NET](#) · [HTML](#), [JavaScript](#) & [CSS](#) · [Jupyter Notebooks](#) · [MS SQL Server](#)